A 50 Gbps 49 mW CMOS Analog Multiplexer for a DAC Bandwidth Tripler

Keisuke Kawahara, Joe Sawada, Takumi Kamo, Yohtaro Umeda, and Kyoya Takano

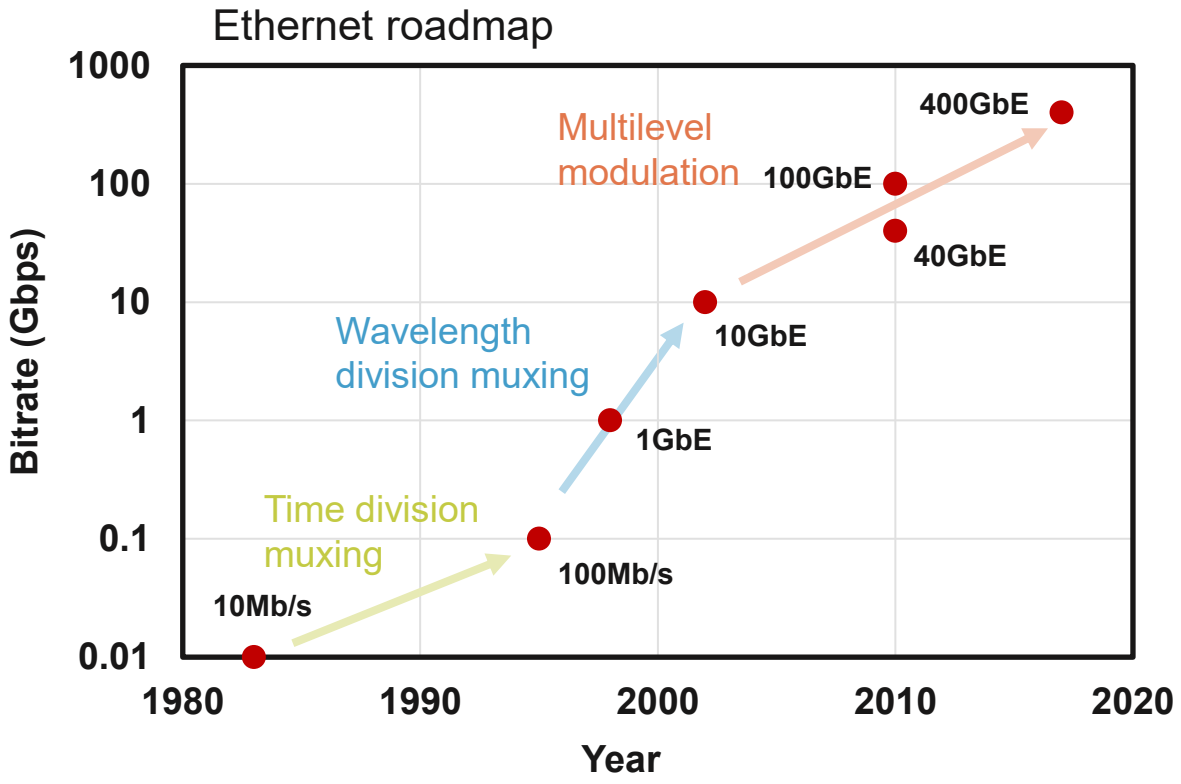
Graduate School of Science and Technology, Tokyo University of Science

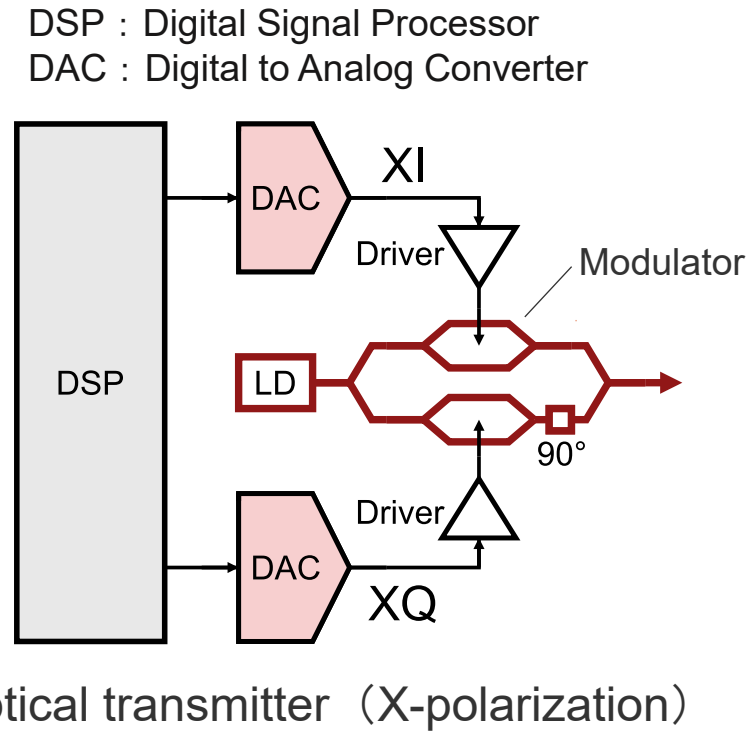
2022 IEEE Radio and Wireless Symposium (RWS 2022)

16-19 / January / 2022, Las Vegas, Nevada, USA

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Fiber-optic communications



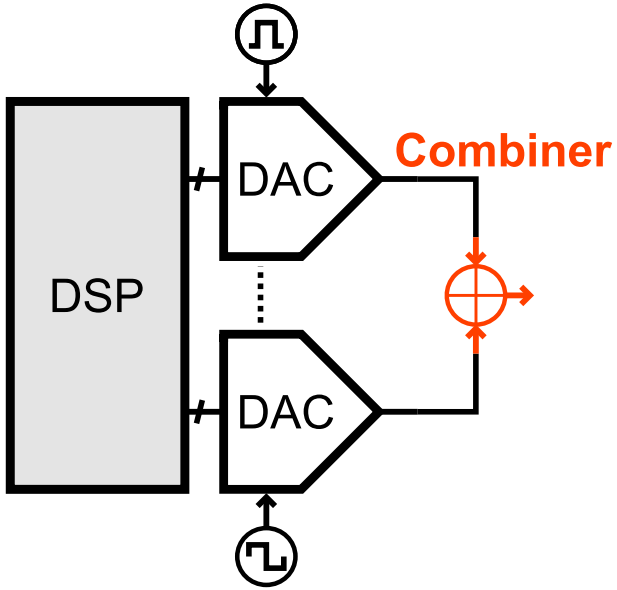


Optical transmitter（X-polarization）

- Multilevel modulation and digital signal processing
  →**High-speed digital-to-analog converters** are required !!

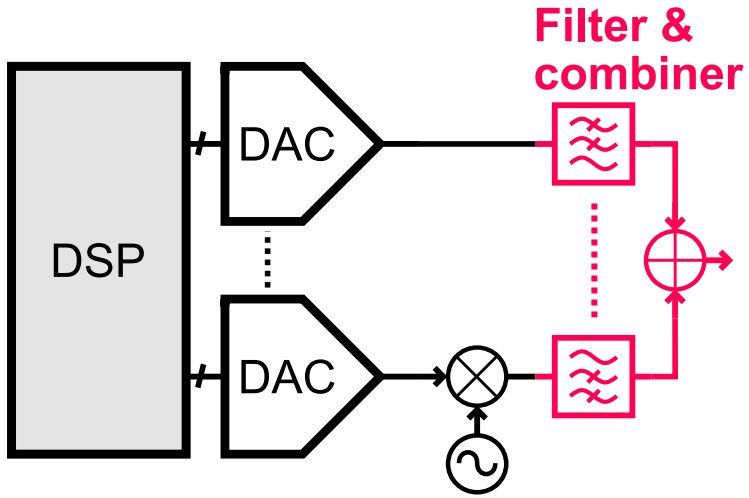
DAC interleaving concepts

Univ. of Stuttgart (2015)
Time interleaving



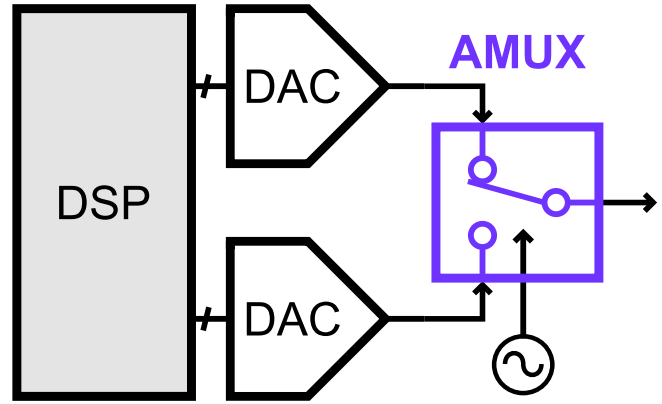
- 😄 Enhances sample rate
- 😥 No bandwidth enhancement

Nokia Bell lab. (2017)
Freq. interleaving



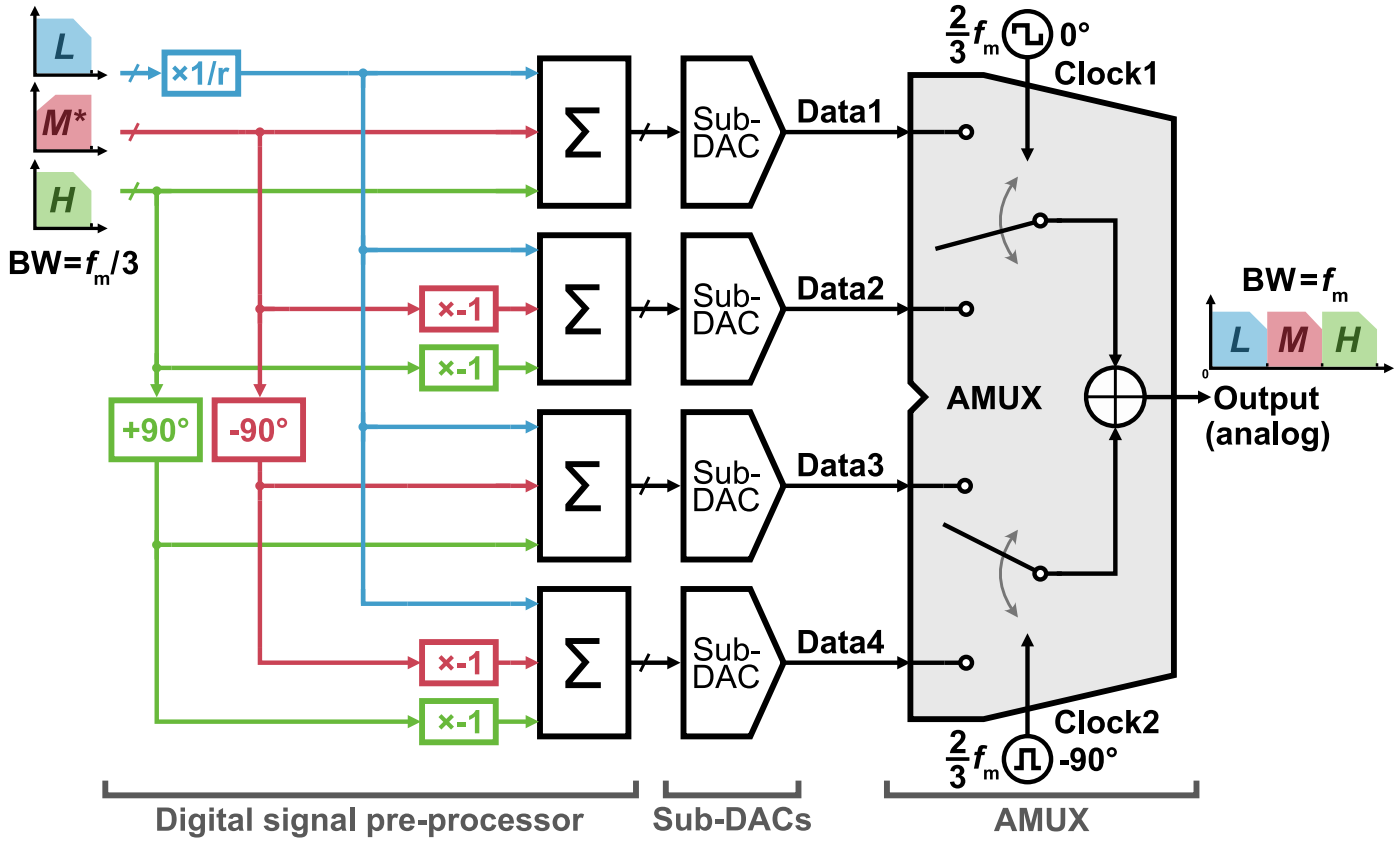
- 😄 Enhances bandwidth by *N*
- 😥 Requires analog filters
- 😥 Delay time mismatch

NTT lab. (2020)
Freq. interleaving
with an analog MUX



- 😄 Not require analog filters
- 😄 No delay time mismatch
- 😥 Bandwidth enhancement is limited up to doubling

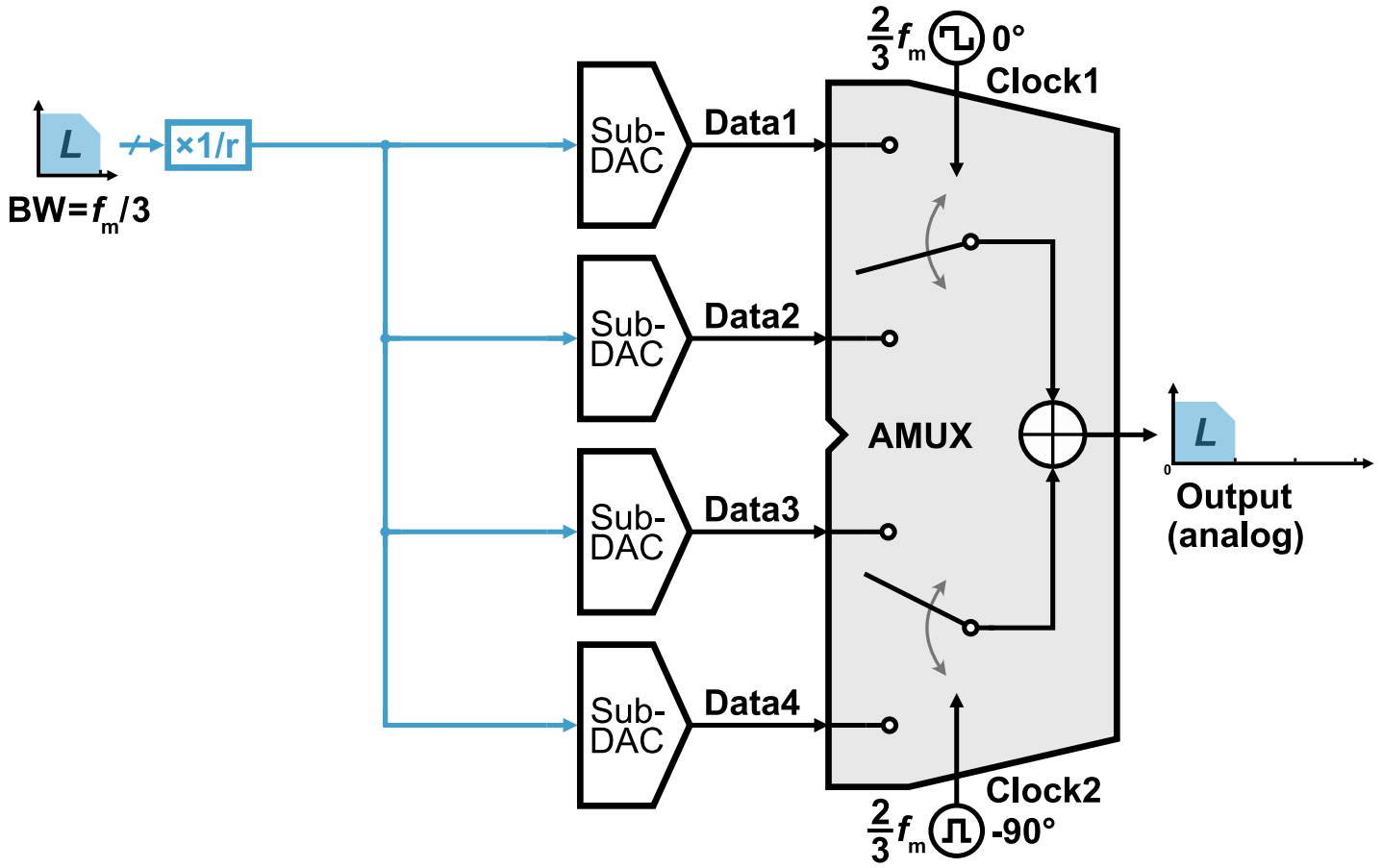
DAC bandwidth tripler



- 😄 No analog filters
- 😄 No delay mismatch
- 😄 Triples bandwidth

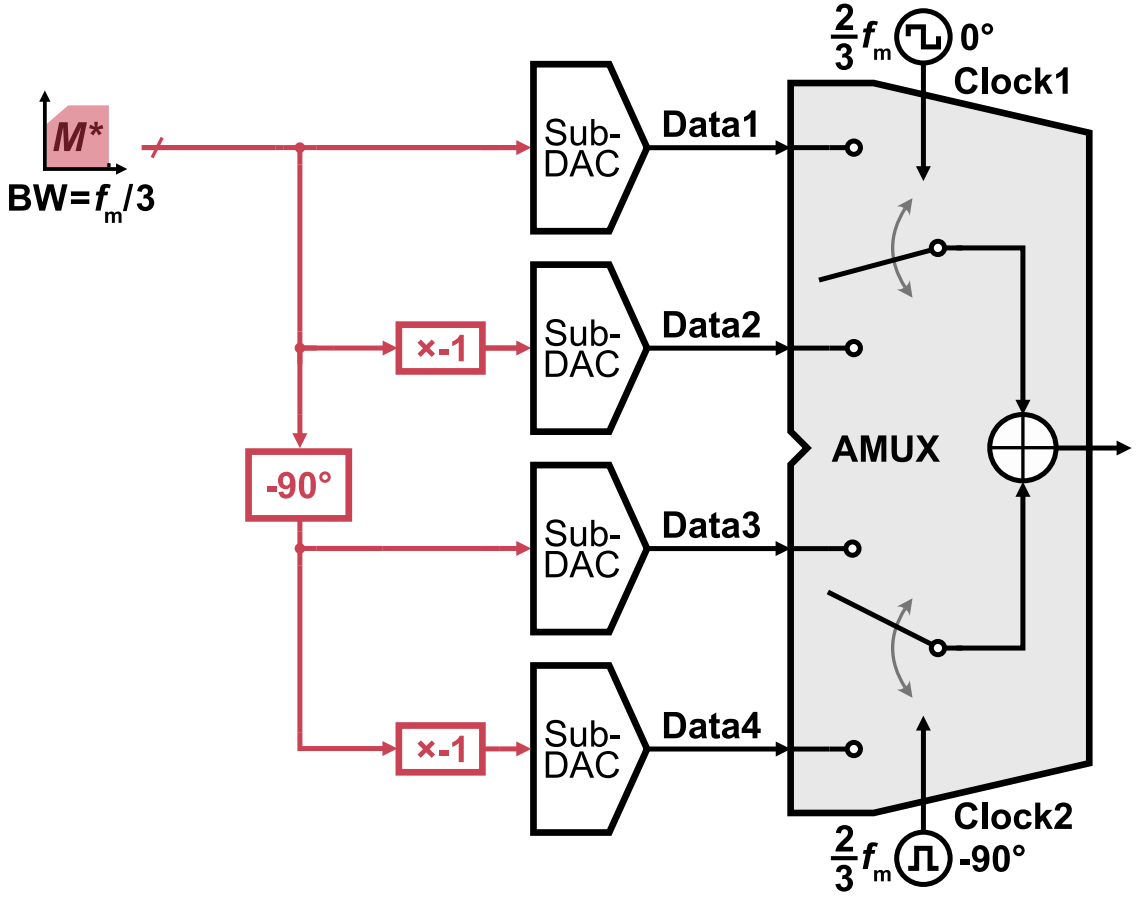
- Our group proposes a “DAC bandwidth tripler”. (Y. Yokoi et al., ICECS 2021)
- Enhances bandwidth by 3 times

Low-band through the AMUX

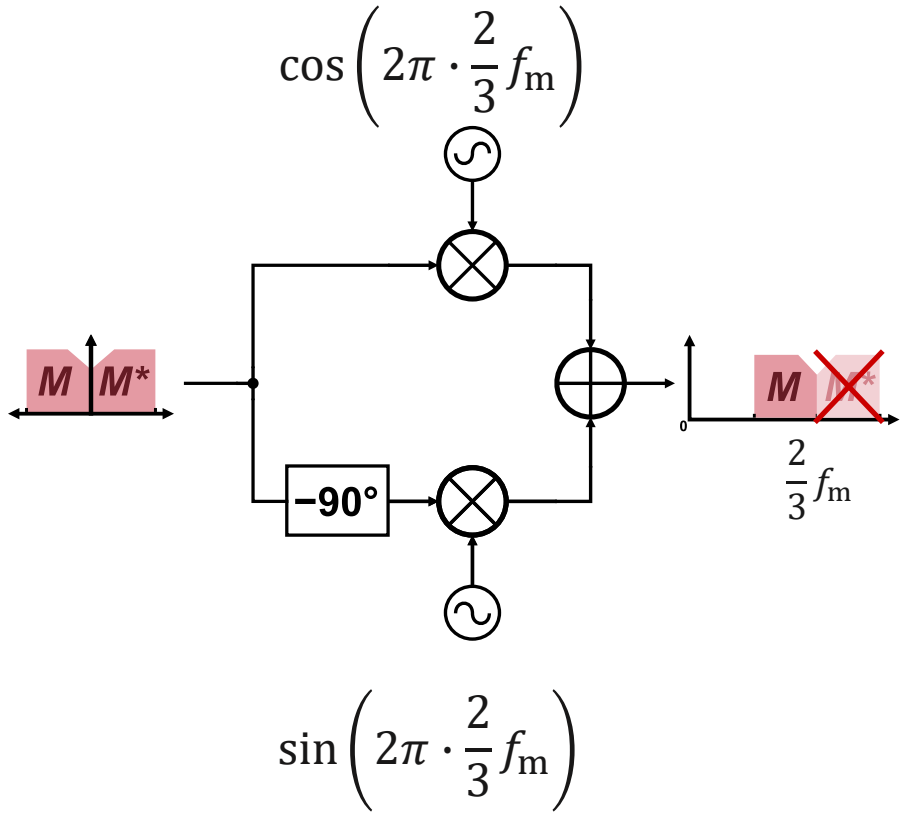


- The same low-band signals are included in all sub-DAC inputs.
- The low-band is output from the AMUX **as it is (i.e., through).**

Mid-band is shifted by 2/3f_m

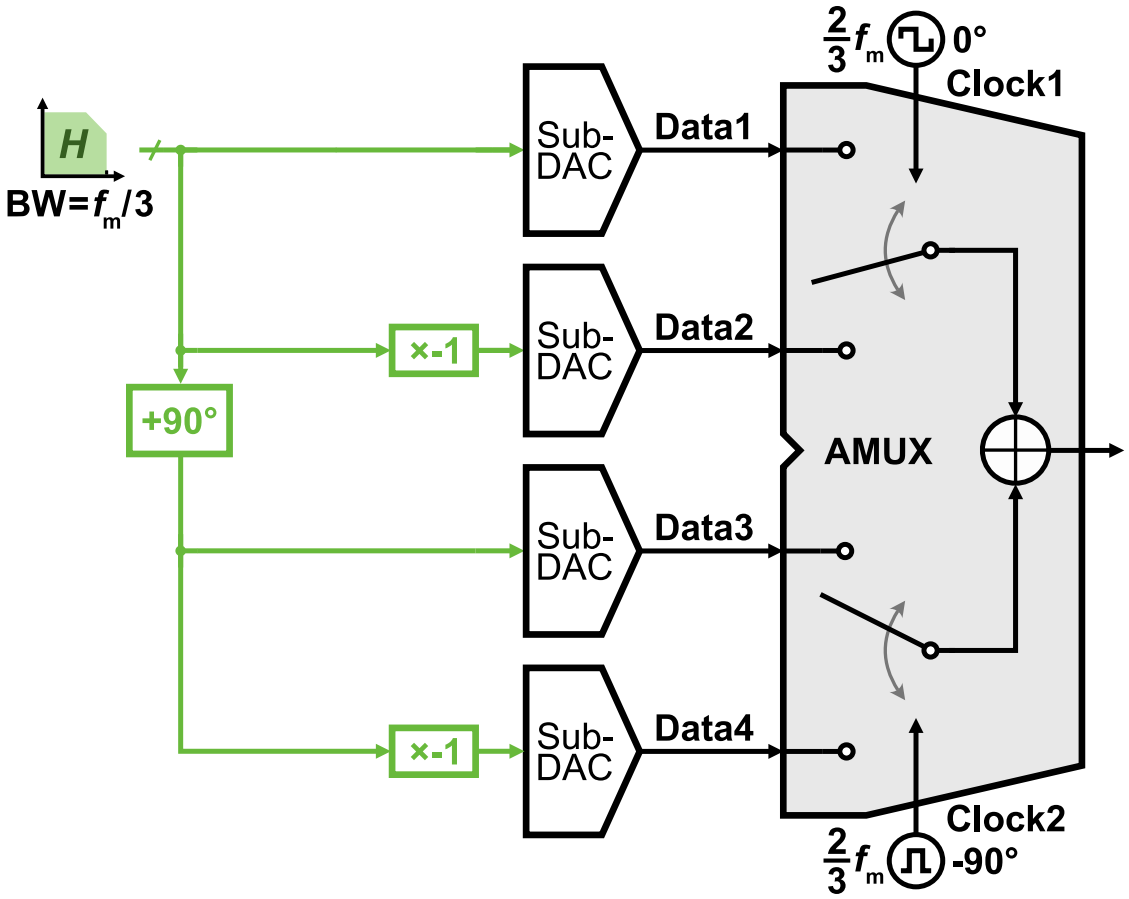


Math. model

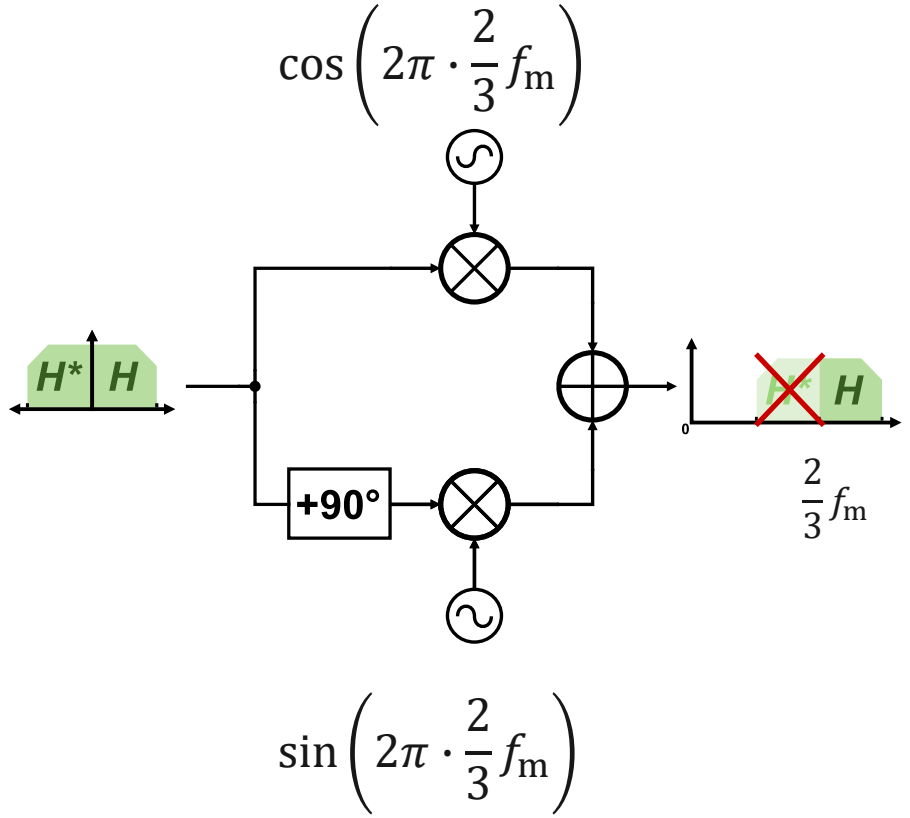


- AMUX operates as a single-sideband (SSB) mixer for the mid-band.
- Only the **lower** sideband is output from the AMUX.

High-band is also shifted by 2/3f_m

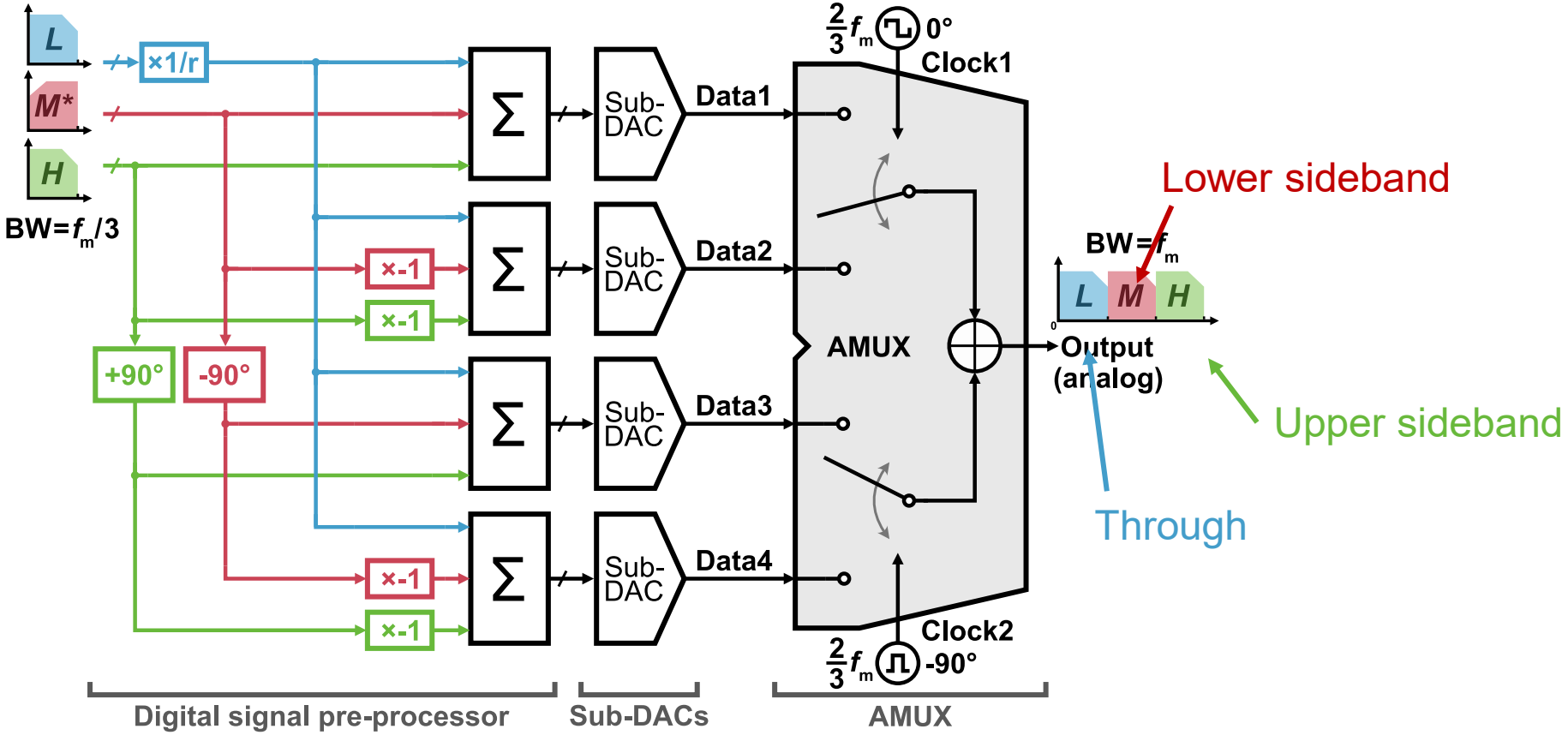


Math. model



- AMUX operates as a single-sideband (SSB) mixer for the high-band.
- Only the **upper** sideband is output from the AMUX.

Whole system again



- Add Low, Mid, and High in advance
- The timing of each sub-band is perfectly aligned.

Challenges

① Development of a **3:1** AMUX-IC

Conventional AMUX-ICs are 2:1 MUX.

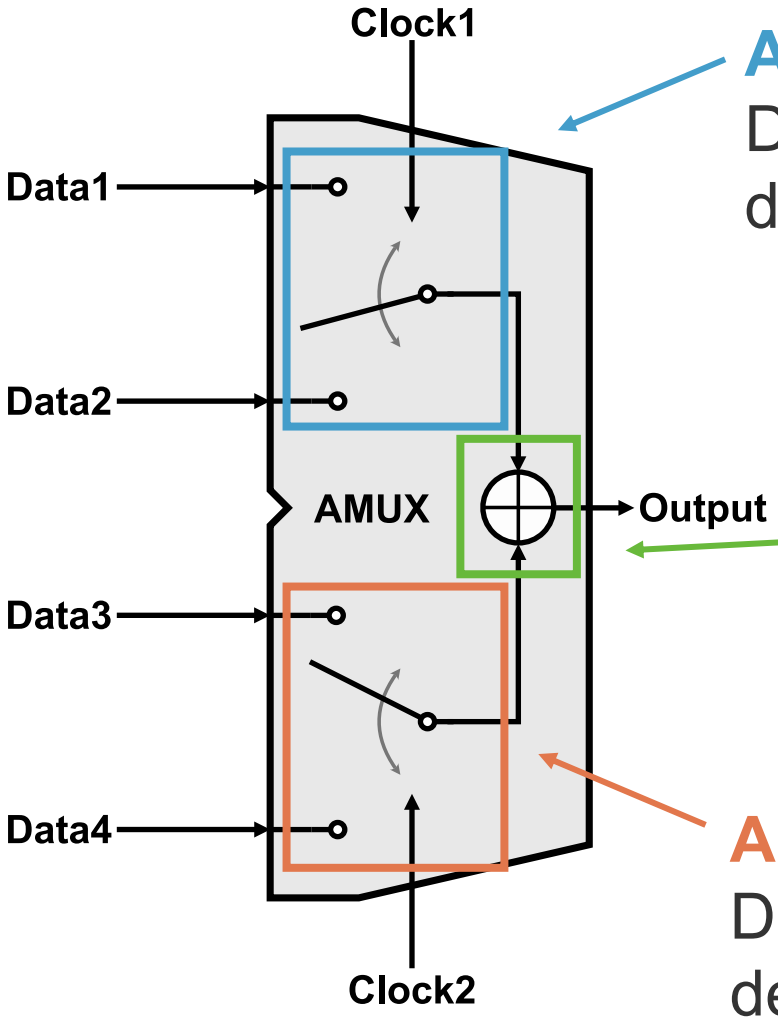
② Reducing power consumption

Conventional AMUX-ICs consume a power of over 500 mW.

To cope with these challenges…

We developed a 3:1 AMUX-IC with a CMOS process in this research.

3:1 AMUX



Alternate switch

Data1 and Data2 are switched depending on the states of ‘High’ or ‘Low’ of Clock1.

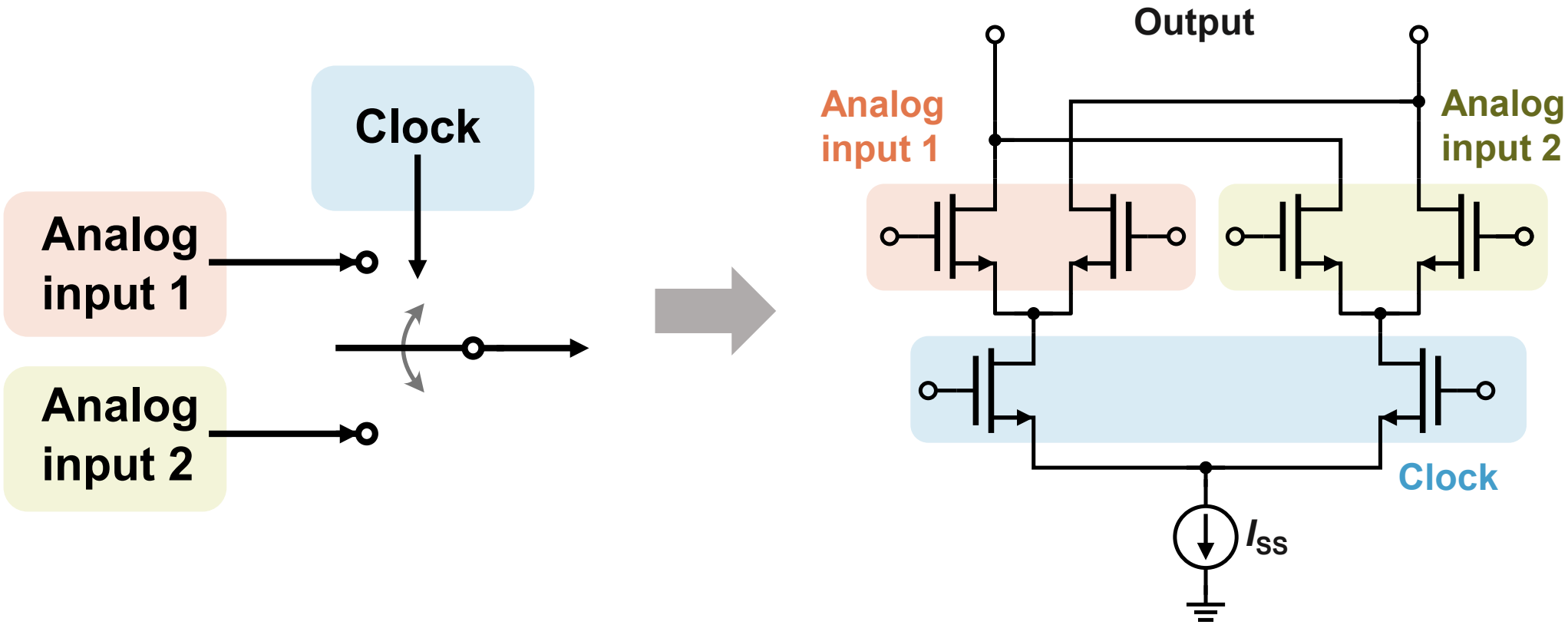
Combiner

The output signals of these switches are added with a combiner.

Alternate switch

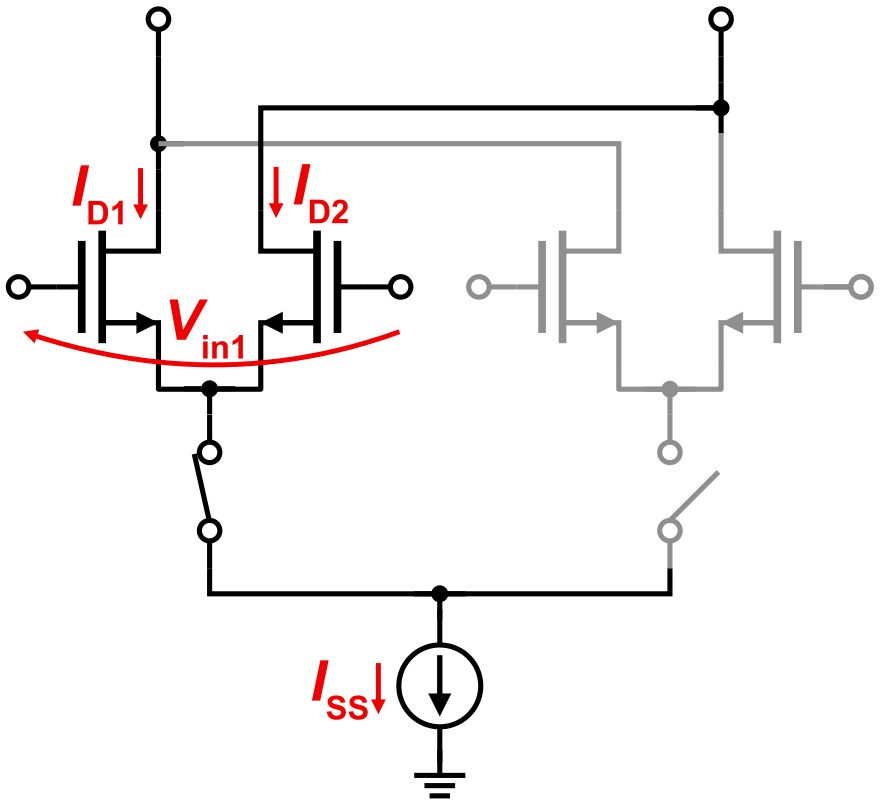
Data3 and Data4 are switched depending on the states of ‘High’ or ‘Low’ of Clock2.

Alternate switch



- Double balanced mixer: ◎High-speed ×Non-linearity
  (Gilbert cell)

Analysis of non-linearity



※β: Gain coefficient of the MOSFET

$$I_{D1} - I_{D2}$$

$$\approx \underbrace{\sqrt{\beta I_{SS}} V_{in1}}_{\text{Fundamental}} - \underbrace{\frac{\beta}{8}\sqrt{\frac{\beta}{I_{SS}}} V_{in1}^3}_{\text{3-rd order distortion}}$$

Ration of the fundamental to the 3rd-order distortion

$$\frac{\frac{\beta}{8}\sqrt{\frac{\beta}{I_{SS}}}}{\sqrt{\beta I_{SS}}} = \frac{\beta}{8I_{SS}} \propto \frac{1}{(V_{GS} - V_{th})^2}$$

→Need to increase $(V_{GS} - V_{th})$ to reduce distortion

Overdrive voltage

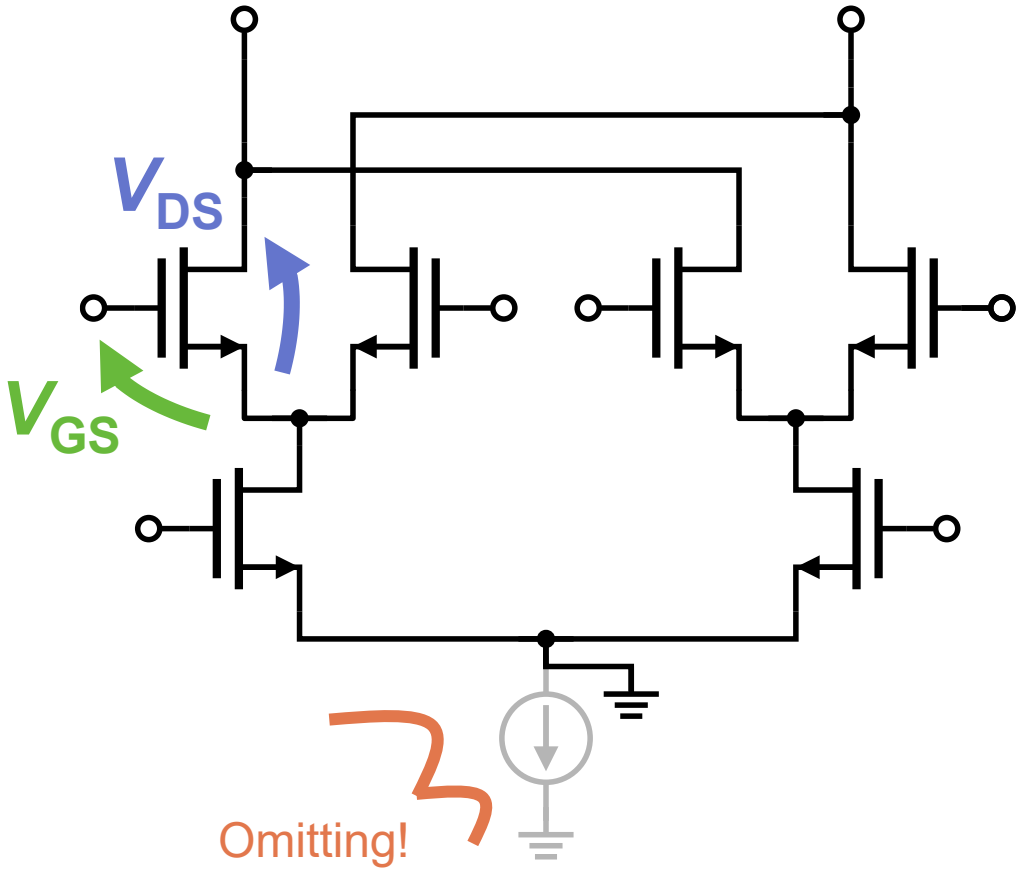
Pseudo-differential switch

For the transistor to operate in the saturation region

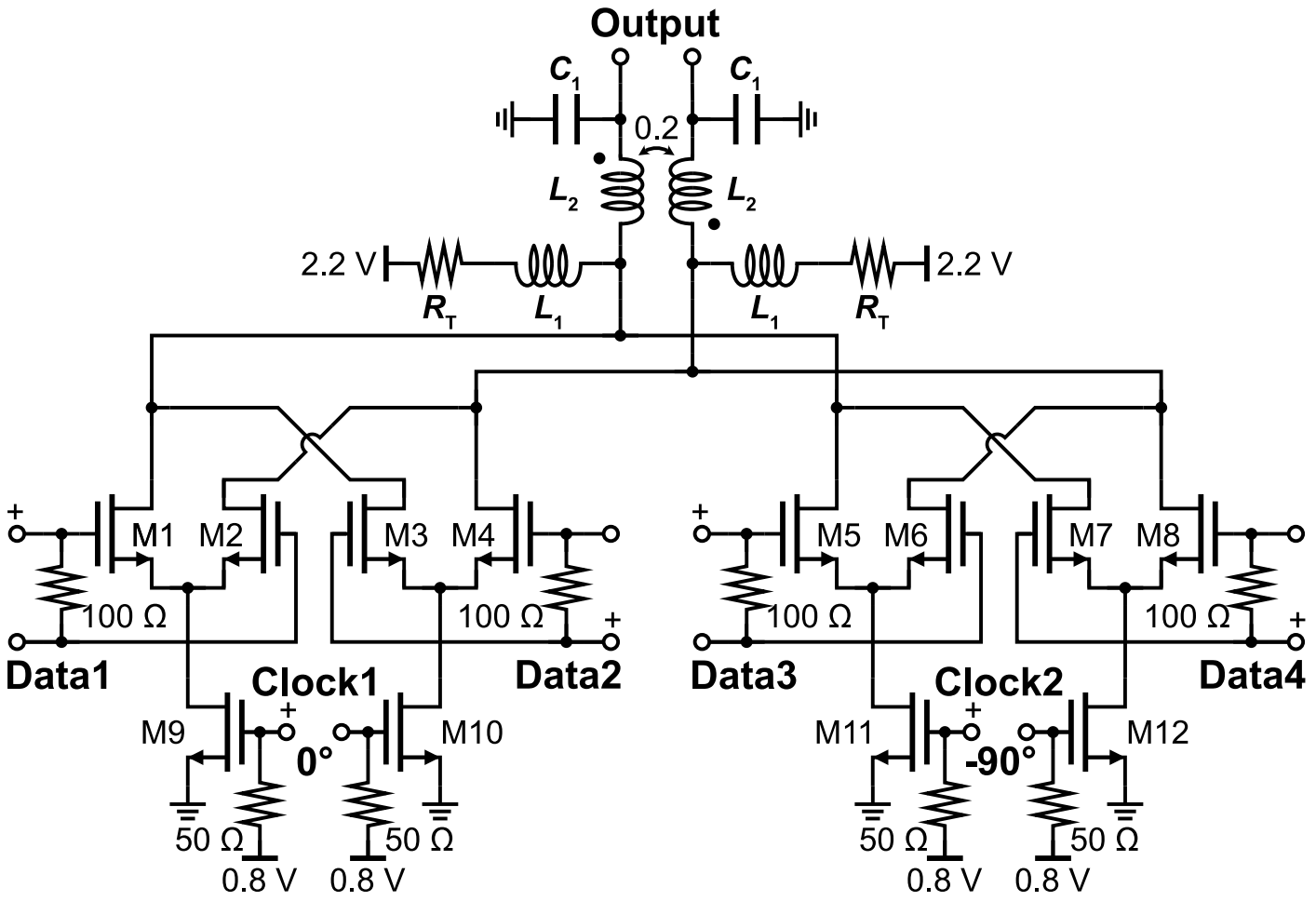
$V_{DS} > V_{GS} - V_{th}$ must be satisfied .

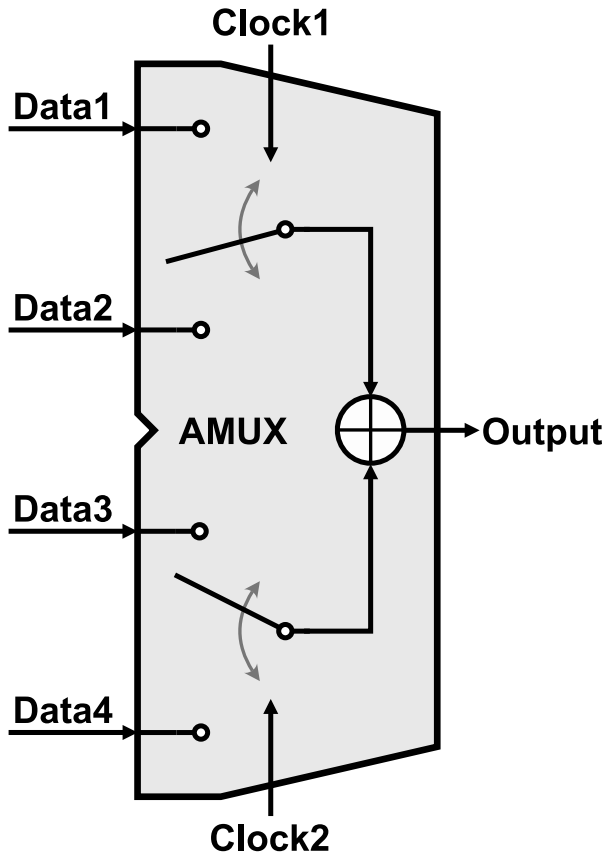
V_{DS} needs to be increased to increase $V_{GS} - V_{th}$

Omitting the current source (Pseudo-differential)

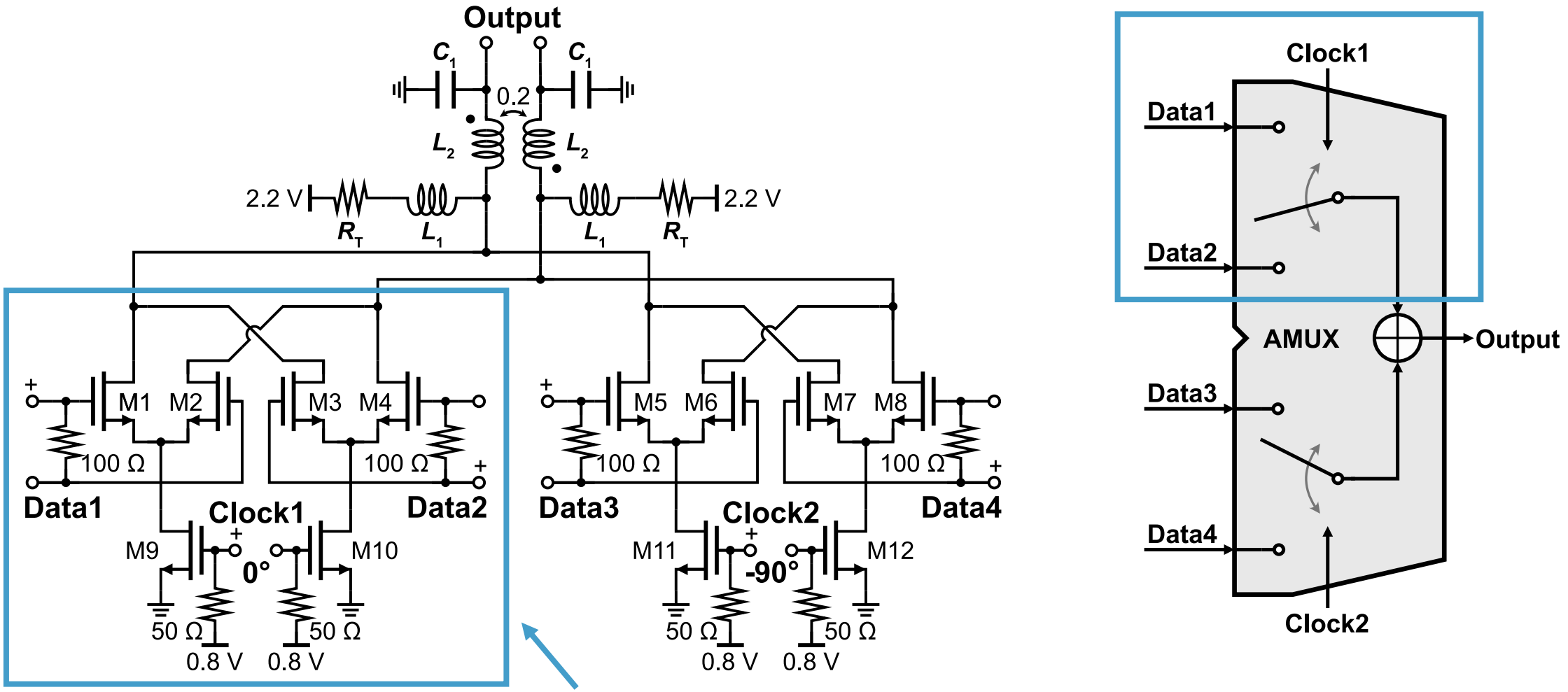


3:1 AMUX (Analog multiplexer)



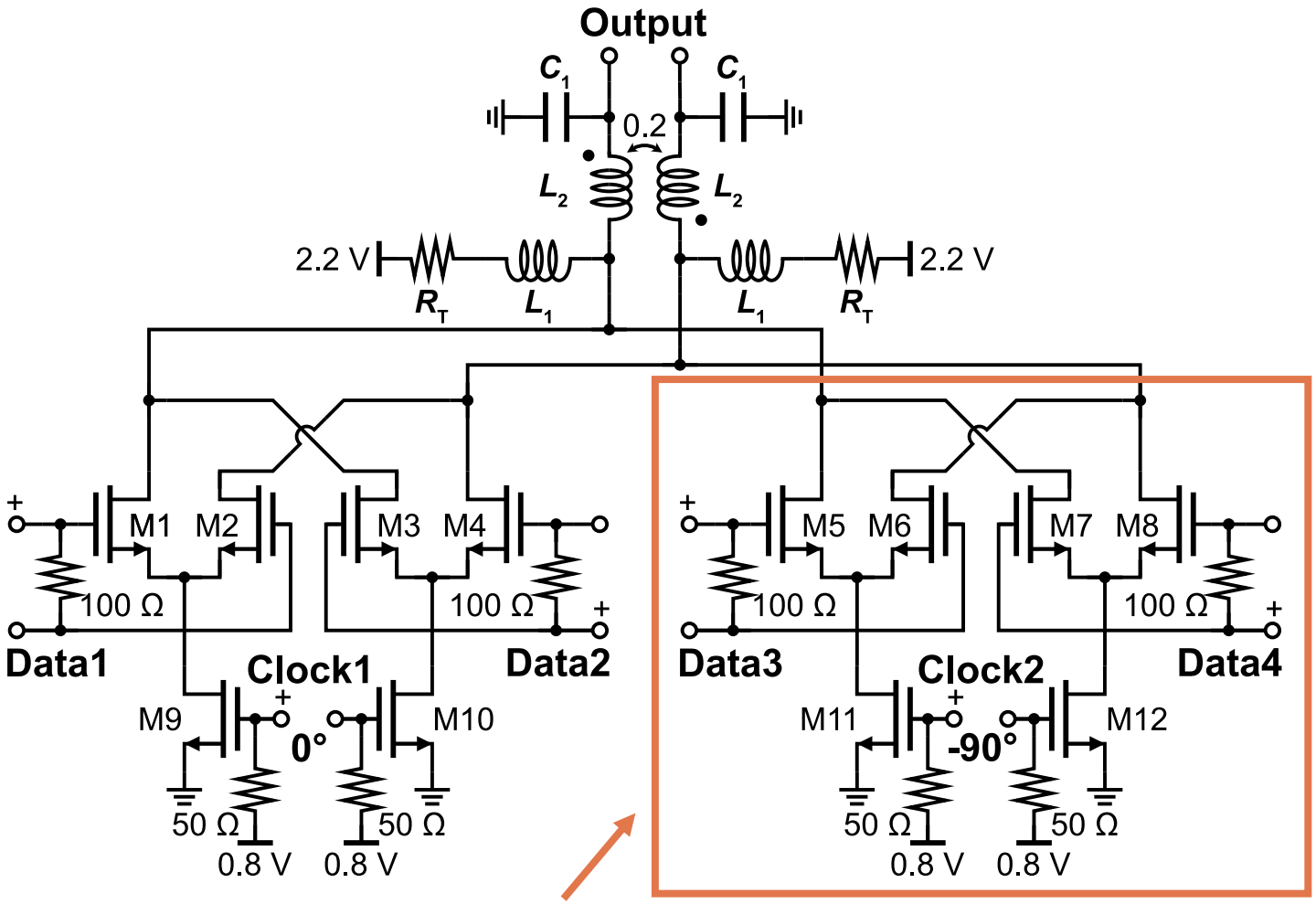


The first switch

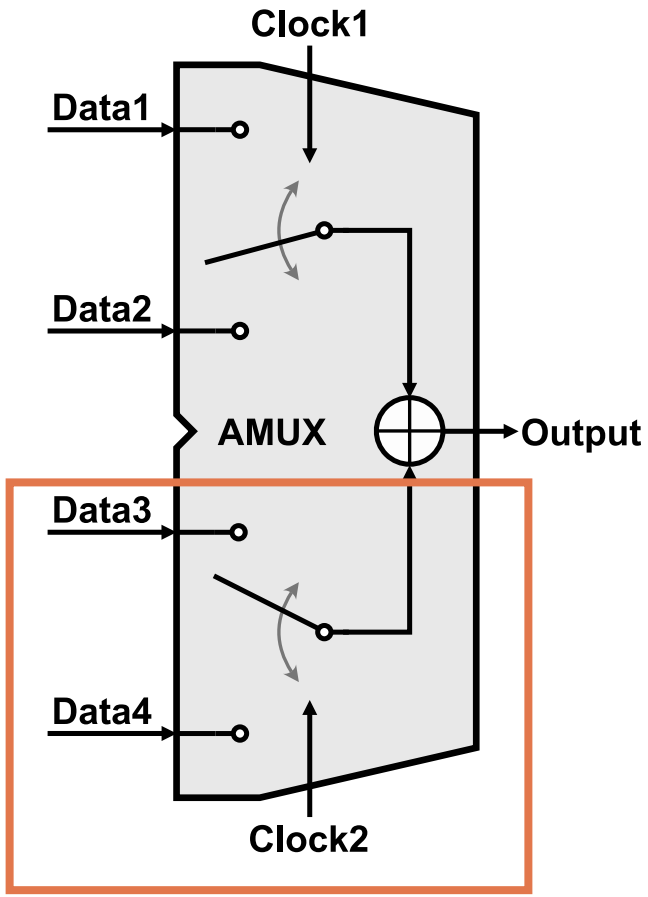


Pseudo differential switch

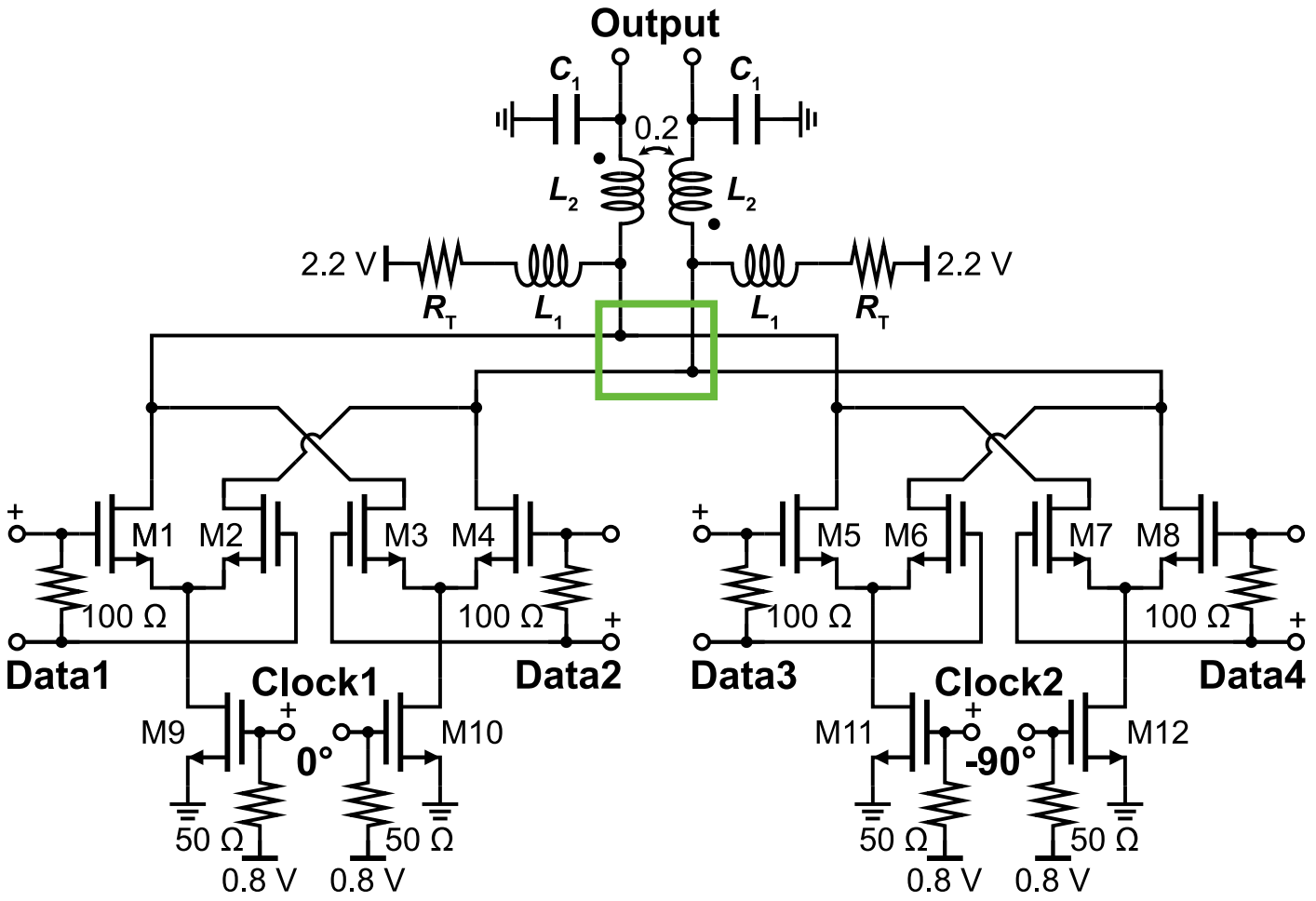
The second switch

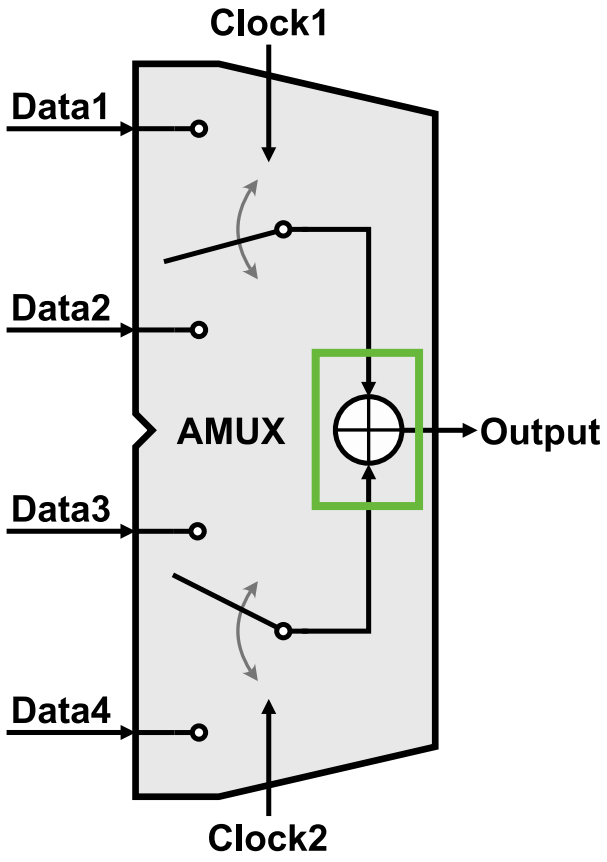


Pseudo differential switch

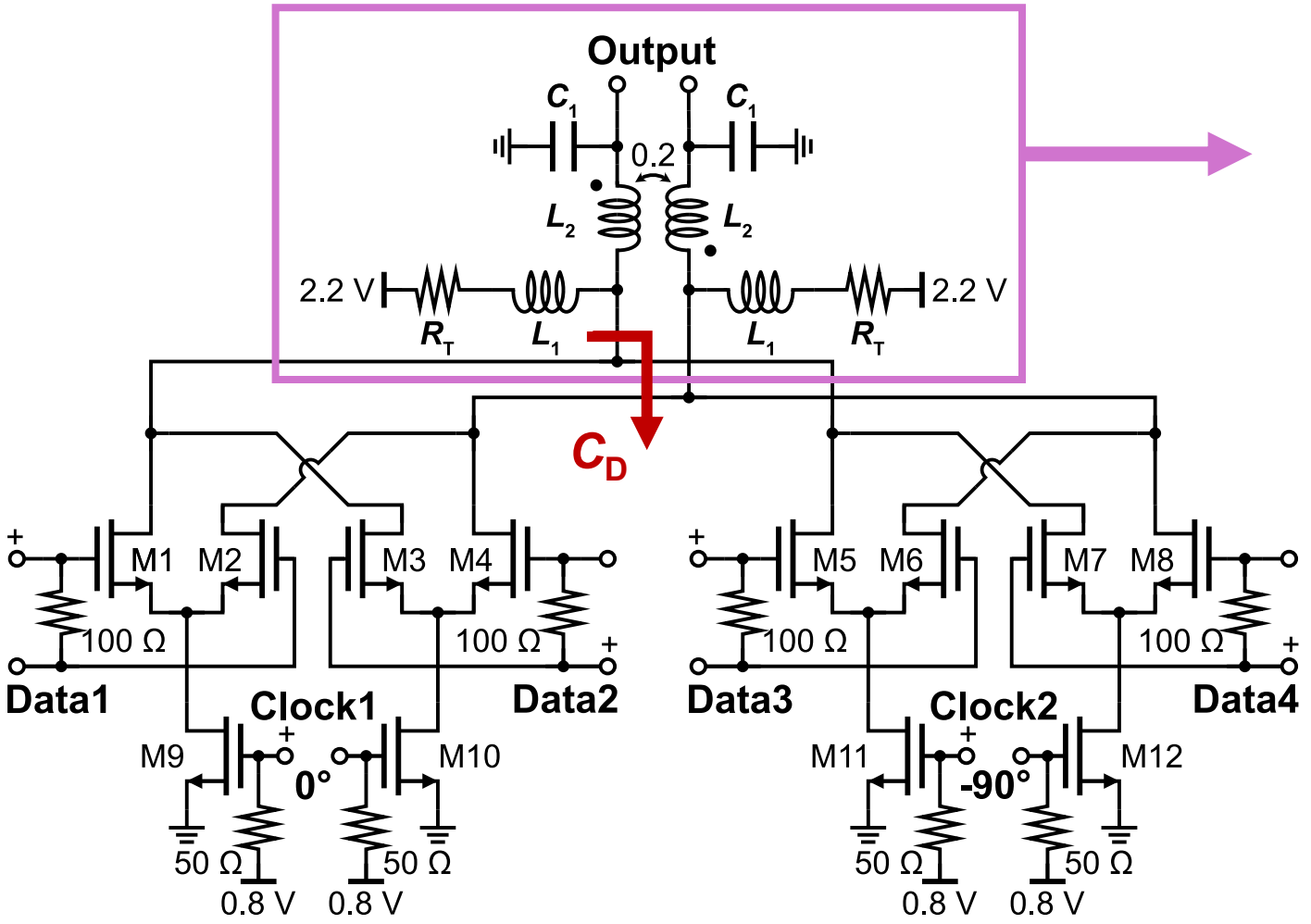


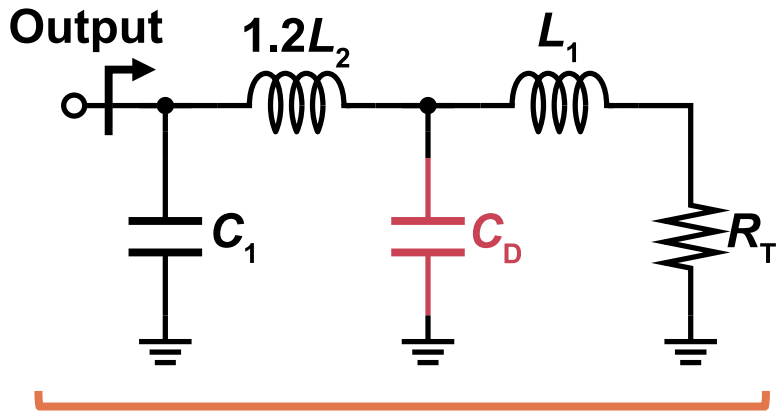
Current mode addition





Broadband termination

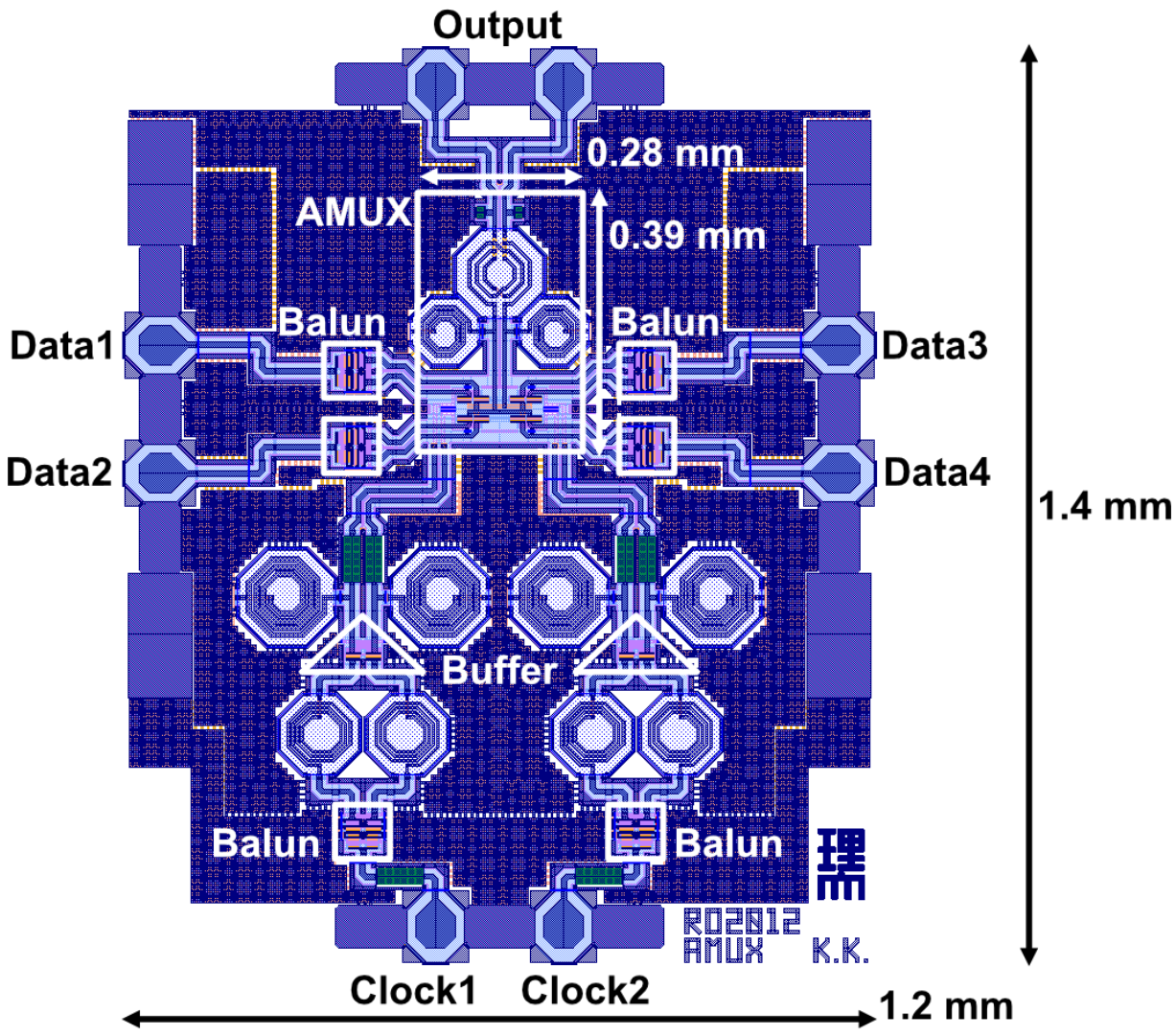




Parasitic capacitor C_D and inductors consist of a **two stage LC ladder**

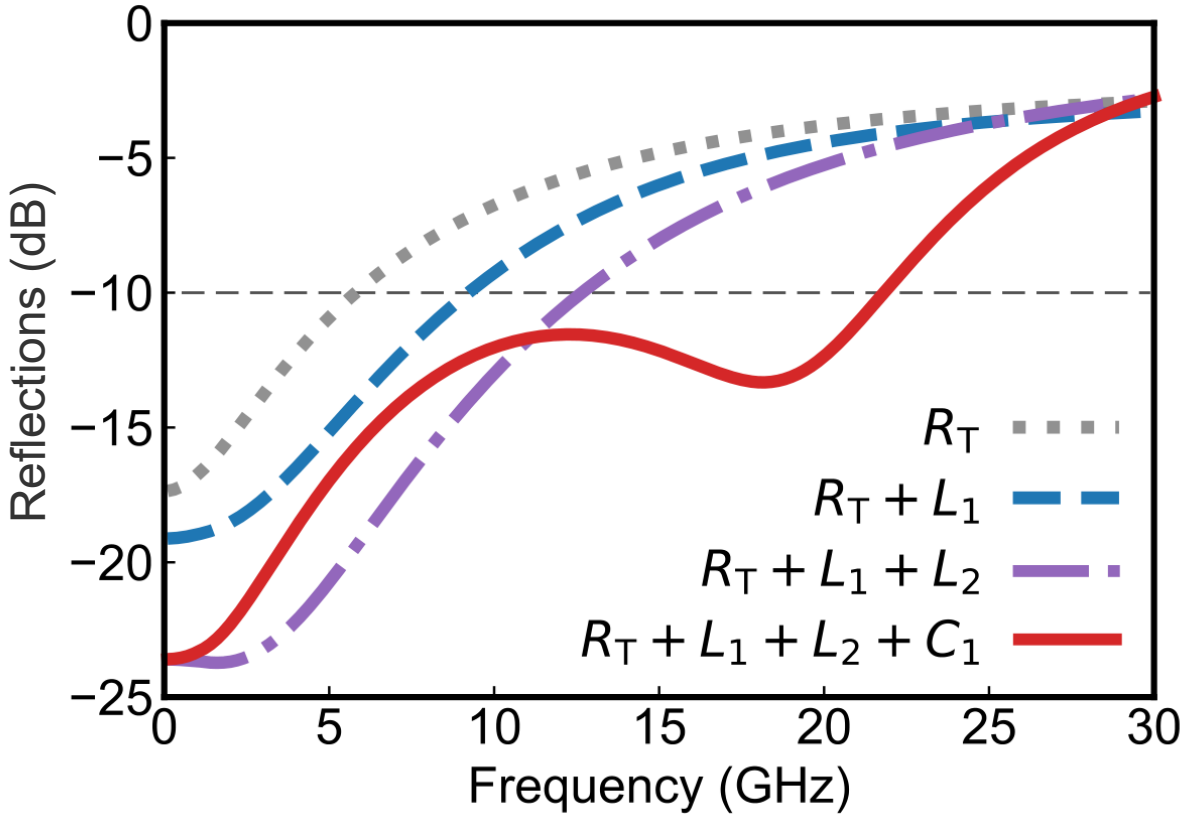
R_T	50 Ω
L_1	450 pH
L_2	300 pH
C_1	130 fF

Layout design



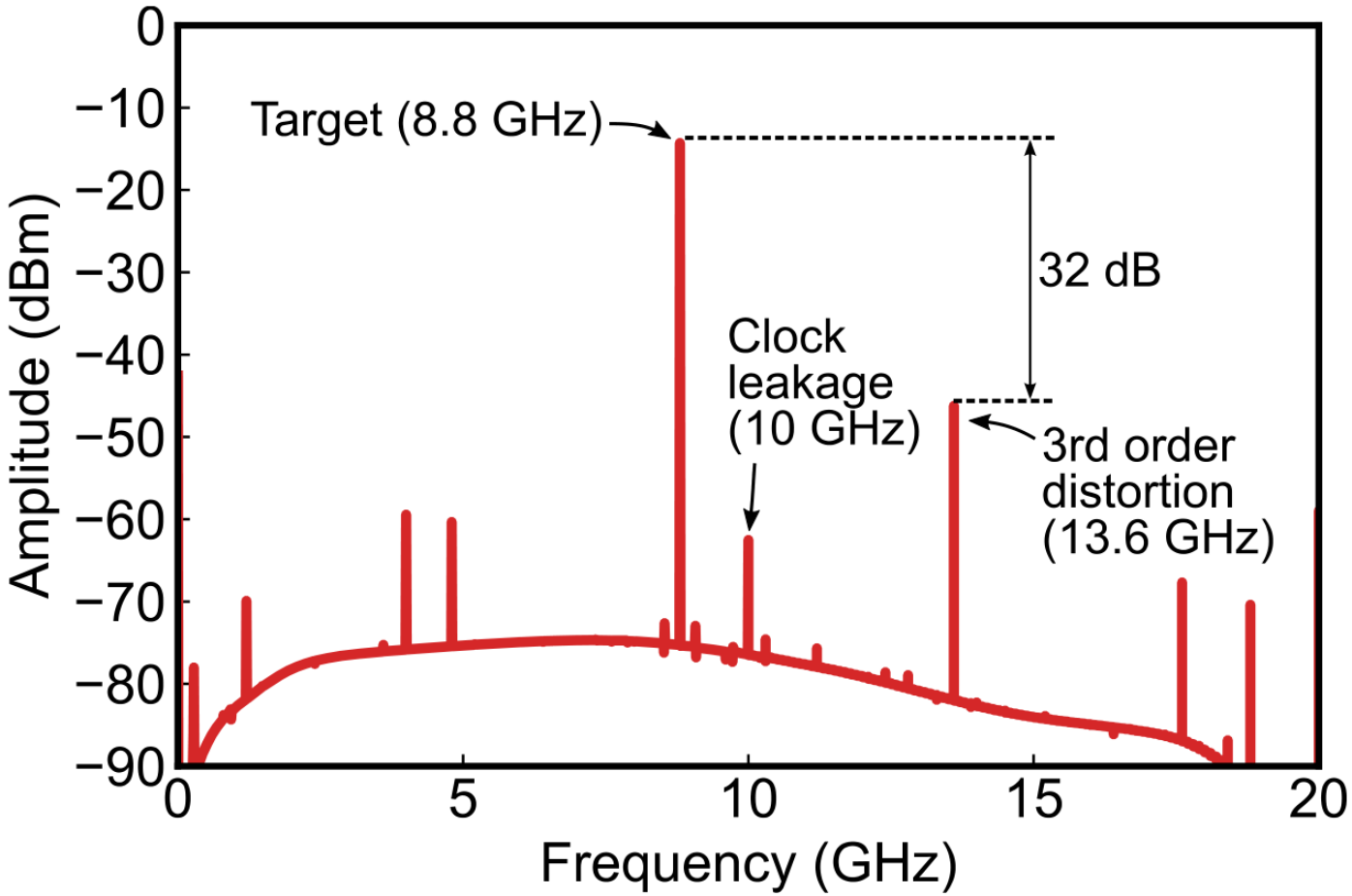
- 180 nm CMOS Technology
- Core area is 0.11 mm^2.
- Baluns are inserted into all inputs to reduce the RF ports.
- Detailed balun design is reported in K. Kawahara et al., MWSCAS 2021.

Output reflections



- Matched bandwidth is 21.4 GHz.
- **3.9x improvement** in matched bandwidth due to L_1 to C_1.

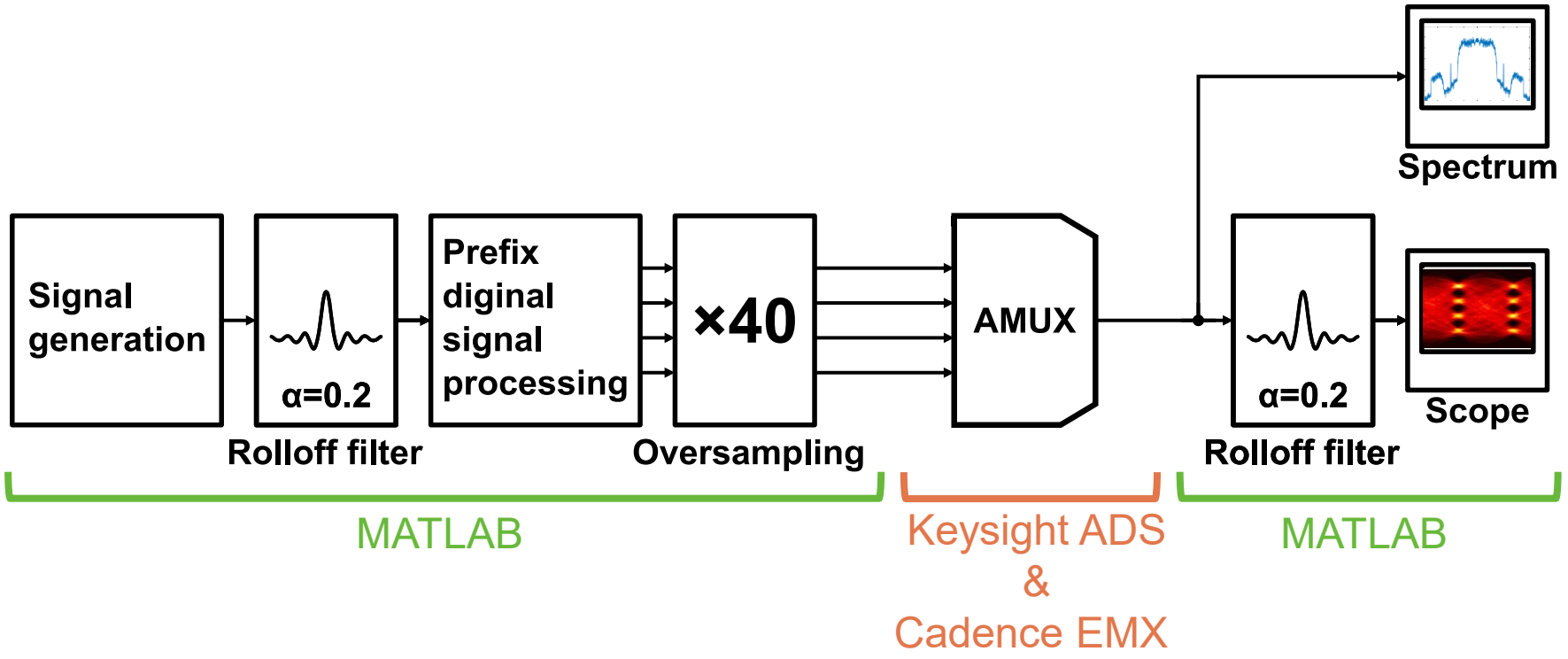
Output spectrum for sine wave



- Data input frequency = 1.2 GHz
- Clock frequency = 10 GHz
- Output lower sideband (8.8 GHz)

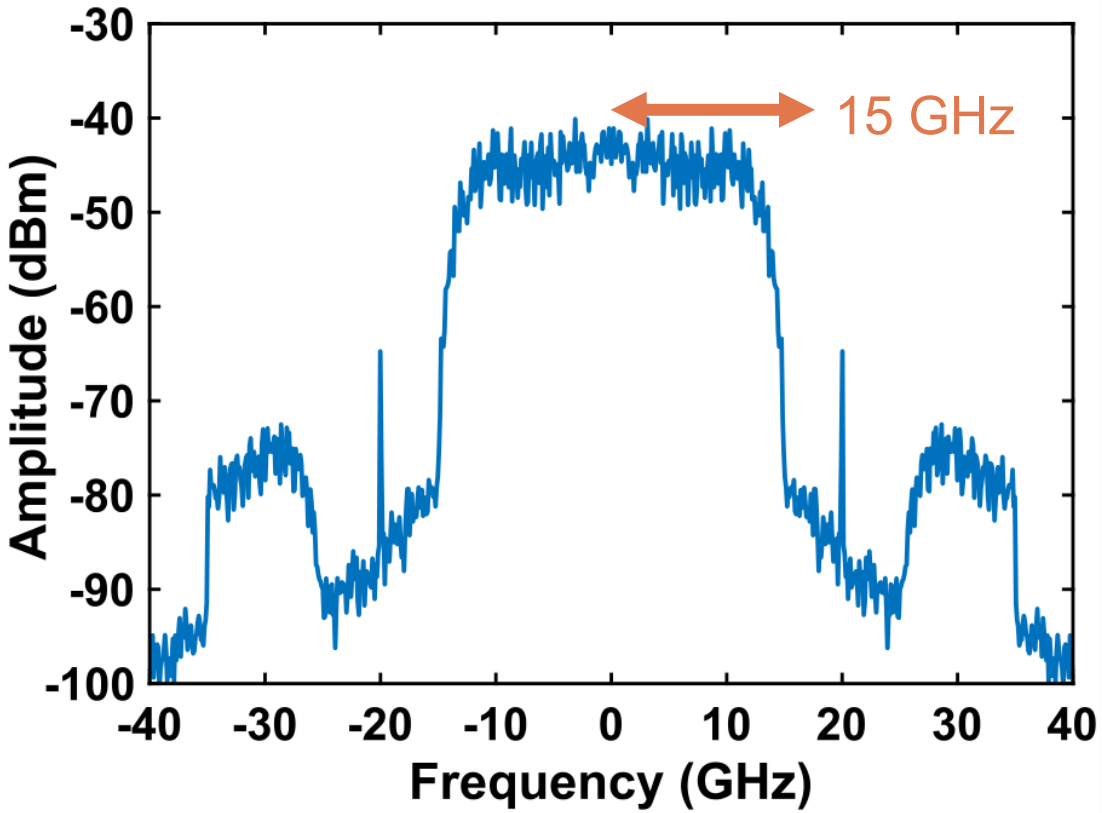
- Satisfy the requirements for PAM4 signal transmission (>30 dB)

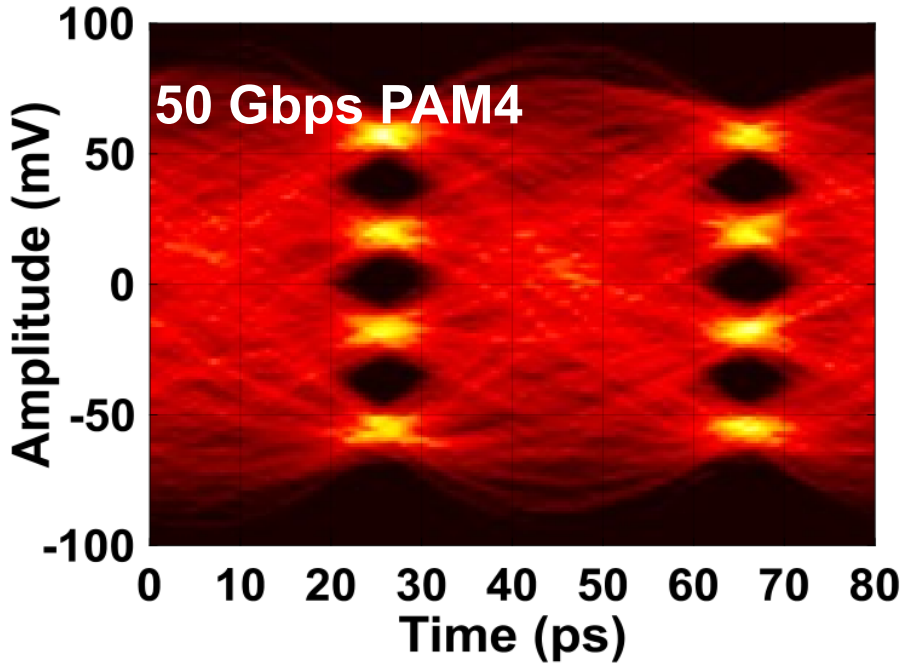
Signal generation test bench



Modulation method	25 Gbaud PAM4
Target bandwidth	15 GHz
Sub-band bandwidth	5 GHz

Signal generation simulation





- Generates a signal with a bandwidth of 15 GHz from an input signal with a bandwidth of 5 GHz
- Power consumption is 49 mW

Comparison with the AMUXes

†Simulation, ‡Excluding the baluns and buffers for measurement

Reference	This work†	[2]	[3]	[4]
Technology	180 nm CMOS	InP HBT	InP HBT	SiGe BiCMOS
fmax (GHz)	65	480	480	500
Multiplication	3:1	2:1	2:1	2:1
Efficiency (pJ/bit)	0.98	4.0	2.9	9.5
−3 dB B.W. (GHz)	22.8	>110	>110	>67
Out. swing (V_{PPD})	0.15	0.5	1.5	0.75
THD (dB)	32	N/A	30	N/A
Baudrate (Gbaud)	25	112	168	56
Baudrate / fmax	0.38	0.23	0.35	0.11
Supply (V)	2.2	-4.5	-4.5	-4.5
Power (mW)	49‡	900	990	1060

- First demonstration of 3:1 AMUX-IC
- Superior power efficiency

Conclusion

- Purposes
  - Realizing high speed digital to analog converters
  - Development of the **3:1** AMUX with **CMOS** process
- Results
  - Low distortion with pseudo differential switches
  - Improved bandwidth with broadband termination
  - 50 Gbps PAM4 signal generation
  - Power efficiency of 0.98 pJ/bit

↪ **Developed techniques improve the bandwidth and power efficiency of optical transmitters !**